# Campus Buddy - MIPS Assembly Project

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**Student ID:** U20230107
**Program:** Personalized Student Life Management System

## Introduction

This report presents the output and functionality of the Campus Buddy system when executed with Student ID verification input **107** (last 3 digits of U20230107). The system demonstrates a complete personalized student life management solution built in MIPS assembly language.

## Algorithms Used for Personalization

### Algorithm 1: Focus Day Calculation

```
Focus Day = Student ID % 6
107 % 6 = 5 → Saturday (Day 5)
```

### Algorithm 2: Campus Personality Identification

```
Personality = (Last Digit >> 2) % 4
(7 >> 2) % 4 = 1 → Sports Enthusiast
```

### Algorithm 3: Optimal Study Time Detection

```
Study Time = (Sum of all digits) % 5
(1+0+2+3+0+1+0+7) % 5 = 0 → Morning
```

### Algorithm 4: Campus Personality Classification

```
Personality Type = (Last Digit of ID >> 2) % 4
Result Mapping:
  0 = Library Scholar
  1 = Sports Enthusiast
  2 = Wellness Advocate
  3 = Social Butterfly
```

### Algorithm 5: Focus Day Selection

```
Focus Day = Student ID % 6
Result Mapping:
  0 = Monday, 1 = Tuesday, 2 = Wednesday
  3 = Thursday, 4 = Friday, 5 = Saturday
```

### Algorithm 6: Study Window Time Blocks

```
Study Window = (Sum of Student ID digits) % 5
Result Mapping:
  0 = Morning (9-11 AM)
  1 = Afternoon (2-4 PM)
  2 = Evening (6-8 PM)
  3 = Late Night (9-11 PM)
  4 = Post-midnight (11 PM-1 AM)
```

### Algorithm 7: Time Remaining Calculation

```
Current Time: 12:00 PM (Saturday)
Time Until Next Class: 2:00 PM - 12:00 PM = 120 minutes
```

## Program Output Screenshots

## Feature 1: Input Verification

Program requests Student ID verification by prompting for last 3 digits. Enter **107** to access the personalized dashboard.

```
=== CAMPUS BUDDY ===
Enter the last 3 digits of your Student ID: 107
```

## Feature 2: Personalization Welcome

Upon successful verification, system displays welcome message with current time and student identity.

```
✓ Welcome! Displaying your personalized dashboard...
Current Time: 12:00 PM (Saturday)
Student: U20230107 | Saturday
```

## Feature 3: Personalized Profile

Shows unique ID-based characteristics: Focus Day (Saturday), Personality (Sports Enthusiast), Study Time (Morning 9-11 AM).

```
========== TODAY'S SCHEDULE ==========

========== TIMETABLE (Saturday) ==========
11:00-12:00 Electromagnetics - 2203
12:00-2:00 LUNCH at Mess
2:00-5:00 ILGC - 2201
```

## Feature 4: Time Remaining Until Next Class

Displays current activity and time remaining until next class. Currently in lunch break with 120 minutes until ILGC at 2 PM.

```
========== TIME REMAINING UNTIL NEXT CLASS ==========
[Currently in LUNCH BREAK - 2 hours (12:00 PM - 2:00 PM)]
Time remaining: 120 minutes
```

## Feature 5: Free Slots & Study Plan

Lists available free time slots throughout the day with recommended study activities and locations.

```
========== FREE SLOTS & STUDY PLAN (Saturday) ==========
8:00-9:00 [1 hour FRESHUP & BREAKFAST]
  → Freshen up at Hostel, then breakfast at Mess

9:00-11:00 [2 hours FREE]
  → STUDY: SMAI (Go to Hostel Room, Study + Freshen up)

12:00-1:00 [1 hour FREE]
  → STUDY: DL or EM (Go to Library)

5:00-5:30
  → Relax
```

## Feature 6: Evening Routine

Recommended schedule after classes balancing academics with sports and social activities.

```
========== EVENING ROUTINE ==========
5:30-6:00 PM: Snack break at Mess
6:00-7:00 PM: Basketball
7:30-9:00 PM: Dinner at Mess
9:00-11:30 PM: Free time for assignments, clubs, friends, or rest
11:30 PM: Recommended sleep time
```

## Feature 7: Academic Overview

Current CGPA (8.18) with target (8.3), credit plan, and maximum achievable CGPA calculation.

```
========== ACADEMIC OVERVIEW ==========
Current CGPA: 8.18000031 | Credits: 102/160
(Semester 5: 24 credits)

Target CGPA: 8.30000019

Credit Plan (Personalized):
Sem 5 Credits: 24 -> Goal SGPA: 9.50000000
Sem 6 Credits: 24 -> Goal SGPA: 9.50000000
Sem 7 Credits: 18 -> Goal SGPA: 9.50000000

Note: With 9.5 SGPA in all remaining semesters, maximum achievable CGPA = 8.11
```

## Feature 8: Attendance Planner

Attendance status for all 6 subjects with percentages and grading weightage information.

```
========== ATTENDANCE PLANNER ==========
Calculation: minimum classes x so that (attended+x)/(total+x) >= 75% cutoff

[FOCS]: 88% attendance -> Good standing! (Grading: 10% weightage for attendance)
[Narrative Twists (NT)]: 20/24 classes (83.33%) -> Good standing! (Grading: 20% weightage for attendance)
[Deep Learning (DL)]: 18/20 classes (90%) -> Excellent! (Grading: 10% weightage for attendance)
[Search Methods in AI (SMAI)]: 25 total classes -> (Grading: No attendance weightage)
[Theory of Computation (ToC)]: 84% attendance -> Good standing! (Grading: 10% weightage for attendance)
[Electromagnetics (EM)]: 23/25 classes (92%) -> Excellent! (Grading: 10% weightage for attendance)
```

## Feature 9: Smart Suggestions & Activity Recommendations

Recommended activities and study suggestions including basketball, club meetings, and relaxation time.

```
========== SMART SUGGESTIONS & ACTIVITY RECOMMENDATIONS ==========
[6:00-7:00 PM] BASKETBALL PRACTICE
[9:00-10:00 PM] GEEK ROOM CLUB MEETING (Monday - Saturday)
[9:00-10:00 PM] ROBOTICS LAB MEETING (Every Sunday)
[10:00-11:00 PM] FREE TIME (Games, Personal Projects, Relaxation)
```

## Feature 10: Subject Difficulty Ratings

Difficulty levels of all 6 subjects (1=Easy, 5=Hardest) to prioritize study time allocation.

```
=== SUBJECT DIFFICULTY RATINGS (1=Easy, 5=Hardest) ===
Search Methods in AI: Difficulty Level 5
Theory of Computation: Difficulty Level 4
Deep Learning: Difficulty Level 3
Electromagnetics: Difficulty Level 3
Fundamentals of CS: Difficulty Level 2
Narrative Twists: Difficulty Level 1
```

## Feature 11: Smart Study Recommendations

Recommended study focus for free slots, prioritizing hardest subjects when mind is fresh in morning.

```
===== SMART STUDY RECOMMENDATIONS (For Free Slots 9 AM - 5 PM) =====
1. SMAI (Hardest) - Study when fresh
2. TOC (2nd Hardest) - After SMAI
3. DL, EM, FOCS - Quick reviews & assignments
4. NT - Last priority
```

## Feature 12: Smart Location Recommendations

Personalized study locations based on personality profile and optimal study time for each part of the day.

```
========== SMART LOCATION RECOMMENDATIONS (Personalized) ==========
Based on your profile and schedule:

[Morning (9-11 AM)] Go to HOSTEL ROOM - Study when your mind is fresh
[Midday (12-1 PM)] Go to LIBRARY - Study Deep Learning (DL) between lunch & ILGC class
[Evening (6-8 PM)] Go to BASKETBALL COURT - Your best time for basketball and activities
[Night (9-11 PM)] Go to HOSTEL and attend GEEK ROOM Meeting
[SATURDAY - Your Priority Day] Use morning free time (9-11 AM) for long study sessions
```

## Technical Details

- **Language:** MIPS Assembly
- **Code Length:** 920 lines
- **Features:** 12 major components
- **Personalization:** 7 algorithms
- **Student ID:** U20230107